Genome Analysis Suggests Reclassification of a Putative *Clostridium botulinum* Strain

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INTRODUCTION

Toxigenic *Clostridium botulinum* strains produce botulinum neurotoxin, the cause of botulism. Because closely related non-toxigenic clostridia can resemble *C. botulinum*, whole-genome approaches such as average nucleotide identity (ANI) are useful for resolving ambiguous genome classifications compared to 16S rRNA alone

OBJECTIVE

- Research Question: **Is SRR29831631 correctly classified as *Clostridium botulinum*, or is it more genomically consistent with *Clostridium sporogenes*?**
- **Approach:** Use a Galaxy-based workflow to assemble, annotate, screen for botulinum neurotoxin genes, and compare SRR29831631 to reference genomes using ANI.

Methods/Workflow

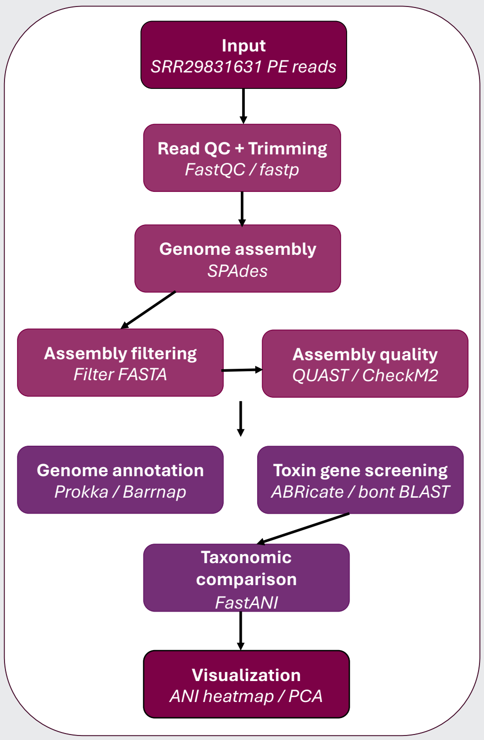

Figure 1. Genome analysis workflow.
Galaxy-based workflow used to quality-check, assemble, annotate, screen, and compare SRR29831631 against reference genomes.

RESULTS

Table 1. Draft genome assembly quality

Metric	Result
Assembly size	3,984,358
Scaffolds ≥500 bp	30
Largest scaffold	1,494,813 bp
N50	746,694
GC content	27.83%
CheckM2 completeness	99.99%
CheckM2 contamination	1.66%

Result Box:

Toxin gene screening

No complete **bont** gene was detected. **Megablast** returned 0 hits, while **blastn** produced only short local alignments rather than full-length toxin-gene evidence

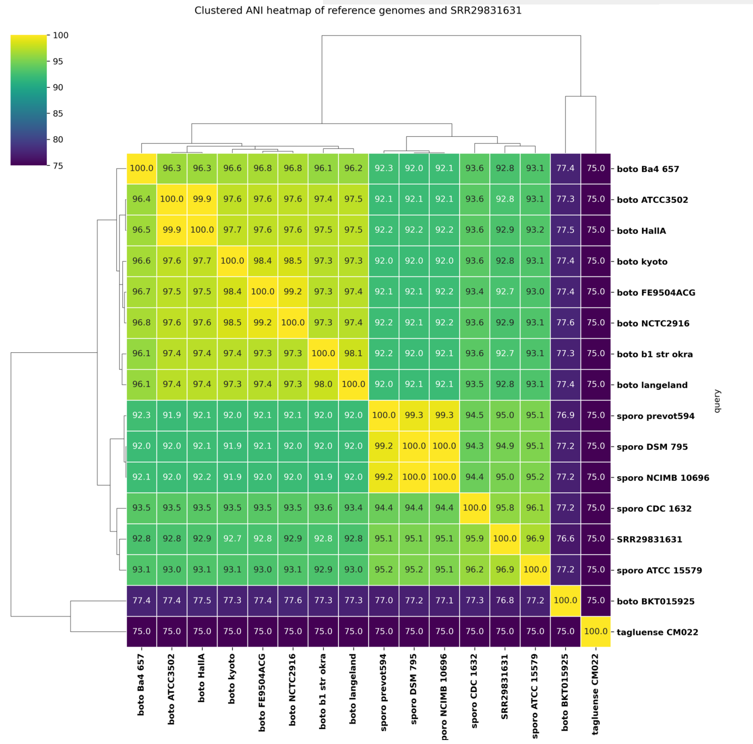

Figure 2. ANI heatmap comparing SRR29831631 to reference genomes.
SRR29831631 showed ≥95% ANI with *C. sporogenes* references but only ~92–93% ANI with *C. botulinum* references, supporting closer genomic similarity to *C. sporogenes*.

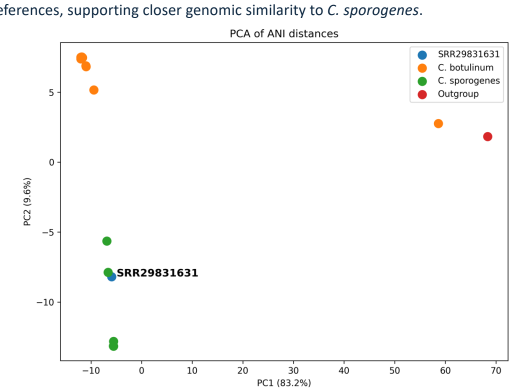

Figure 3. PCA of ANI distances.
Principal component analysis of all-vs-all ANI distances placed SRR29831631 with the *C. sporogenes* reference group rather than the *C. botulinum* cluster.

CONCLUSION

- SRR29831631 assembled into a high-quality draft genome with high completeness and low contamination.
- No complete bont gene was detected by annotation or targeted toxin-gene screening
- ANI heatmap and PCA results support SRR29831631 as a non-toxigenic *Clostridium sporogenes*-like genome rather than toxigenic *C. botulinum*

FUTURE WORK

- Compare SRR29831631 with a larger *C. botulinum/C. sporogenes* reference panel.
- Further investigate AMR-associated CARD/RGI hits with phenotype or comparative genomic analysis.

ACKNOWLEDGEMENTS

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Scan for GitHub repository, workflow files, and References